MUHAMMAD MANSOOR

Unreal Engine Developer
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PROFILE

Unreal Engine Developer with 3+ years of experience designing and engineering scalable multiplayer and AI-driven systems in Unreal Engine 5 (C++ & Blueprints). Specialized in network replication, gameplay architecture, combat systems, and performance optimization across PC and standalone VR platforms. Experienced in building production-ready systems with a focus on scalability, maintainability, and player experience.

CORE TECHNICAL EXPERTISE

Engine & Programming: Unreal Engine 5/4, C++, Blueprint Architecture
Gameplay Systems: Multiplayer Replication, RPCs, Dedicated Servers, Combat Systems, Save Systems, Session Management
AI Systems: Behavior Trees, EQS, Custom Tasks/Services, Reactive Combat AI
Animation Systems: Control Rig, Animation Blueprints, IK Systems, Layered Blending
Physics & Simulation: Chaos Physics, Destruction Systems, Ragdoll
Performance Optimization: Rendering Profiling, Network Optimization, VR Performance Tuning.
Tools & Workflow: Git, Perforce, Visual Studio, Jira, SourceTree.

WORK EXPERIENCE

Unreal Engine Developer October 2024 – Present
Mimar Studios, Islamabad, Pakistan

- Architected and maintained scalable multiplayer gameplay systems supporting **LAN and online sessions for 30+ concurrent players** using Unreal replication and RPC frameworks.
- Designed modular AI combat frameworks using **Behavior Trees and EQS**, supporting dynamic state-driven enemy and vehicle behaviors.
- Developed extensible animation systems leveraging **Control Rig, IK, and layered Animation Blueprints** to support responsive combat and character movement.
- Built high-fidelity VR combat mechanics including weapon handling, recoil simulation, reload systems, and hardware-synced feedback loops.
- Profiled and optimized rendering, physics, and network performance, **maintaining 72–90 FPS on Meta Quest 3 and reducing draw calls by ~30%.**
- Collaborated within a **cross-functional team of 6–8 developers**, delivering gameplay features across iterative production sprints.

Unreal Engine Developer June 2023 – October 2024
Algoryte, Islamabad, Pakistan

- Led development of an interactive VR vehicle configurator featuring **real-time material, mesh, and environment customization** on Meta Quest 2.
- Implemented gameplay interaction systems within pixel-streamed Unreal environments, enabling **low-latency remote user interaction.**
- Designed AR gameplay systems including **spatial detection, real-time spawning, progression tracking, and leaderboard infrastructure.**
- Integrated backend APIs for **persistent player progression and live data synchronization** across sessions.
- Optimized assets and rendering pipelines to **maintain stable 72 FPS on standalone VR hardware** under performance constraints.

Unreal Engine Developer January 2024 – Present
Freelance Projects

- **Architectural Visualization Project (Unreal Engine):** Developed an interactive ArchViz experience allowing real-time customization of room designs, including sofa, bed, washroom tiles, kitchen cabinets and flooring.
- **VR Multiplayer Zombie Game:** Implemented multiplayer functionality with networking, designed and programmed zombie AI using Behavior Trees, and integrated shooting mechanics for immersive gameplay.
- **Meditation VR App (Meta Quest 3):** Created a calming VR environment with spatial sound design, interactive scenes, and smooth performance optimization for a relaxing user experience.
- **VR Paper Throw Game:** Designed and developed a physics-based VR paper toss game with a real-time scoring system and responsive object interaction.
- **Hijrah Journey VR Experience:** Built a virtual journey through Makkah and Medina using Unreal Engine Sequencer, delivering cinematic storytelling and realistic environments.

EDUCATION

Bachelors of Software Engineering June 2023
Foundation University, Islamabad, Pakistan